

Bayesian Techniques in Macroeconomics

Université Paris Dauphine – PSL
Master in Quantitative Economics, second year

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CHAPTER 1 Modern Business Cycle Theory

Content of the Handout

1	What is a business cycle?	p. 3
2	An illustration of a business cycle model through the consumption-saving problem.....	p. 7
	Exercise 1	p. 12
3	Approximation methods	p. 14
	Exercise 2	p. 17

HANDOUT OBJECTIVES

- Understanding the concept of a business cycle;
- Extracting the business cycle component from time series;
- Solving a simple rational expectation model;
- Linearizing a system of equations;

Session roadmap

Economic intuition. Business cycles combine shocks with propagation mechanisms. **Econometric intuition.** Detrending determines which frequencies the model must explain. **Model and data.** We

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map quarterly GDP into a cyclical component and then into a linear expectations system. **MATLAB/Dynare exercise.** Reproduce the filters, solve the toy model, and interpret impulse responses. **Takeaway.** Bayesian analysis begins with an explicit frequency convention and transparent model–data mapping.

1. What is a business cycle?

1.1 A brief history

Back in the 1930s, the emergence of national accounting allowed economists to get an accurate measure of aggregate production from the demand, supply, and income sides. National accounts provide a complete and consistent set of techniques to measure a country's economic activity. Given these new measures of economic activity, the study of business cycles quickly emerged. As explained by [Schumpeter \[1930\]](#), the pioneer of the concept of business cycles can be traced back to [Mitchell \[1913\]](#)'s work on the interpretation of gold, prices, and wages during the US Civil War in his treatise on *Business Cycles*. He provided the first quantitative study of business cycles. His pioneering work laid the foundation for modern business cycle analysis in [Burns and Mitchell \[1946\]](#) by defining a business cycle as “a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; in duration, business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar characteristics with amplitudes approximating their own.”

The contributions of Mitchell established the first dichotomy in economics by separating long-term and short-term analyses in macroeconomics. In particular, [Burns and Mitchell \[1946\]](#) provided the first definition of the duration of a business cycle, which lies between one and twelve years. This definition of the business cycle has evolved over time; modern business cycle analysis determines the frequency of a business cycle as a fluctuation lying between 6 to 32 quarters, which is shorter than the duration suggested by [Burns and Mitchell \[1946\]](#). Any component of a macroeconomic time series with a Components with periods between 6 and 32 quarters are retained as business-cycle fluctuations; components with periods longer than 32 quarters are assigned to the trend.

A clear typology of an economic cycle is a primary aspect of macroeconomic analysis, particularly for policymakers. Consider a government facing an adverse event (financial turmoil, pandemic, war, etc.); dating the recession, gauging its magnitude and duration are all crucial for determining the size of any fiscal/monetary stimulus to mitigate the crisis.² During the last century, economic associations, exemplified by the NBER and CEPR, were established to provide start and end dates of recessions for the US and Europe, respectively. Therefore, significant efforts have been made by economists to clearly typify the economic cycle by disentangling periods of expansion, crisis, recession, and recovery in macroeconomic time series.

Prior to the 1940s, much of the economic cycle analysis focused on determining the periodicity of cycles. You may have heard during your undergraduate studies about the Juglar, Kuznets, and Kondratiev cycles. These cycle analyses aim to explain how long an innovation, investment, or other spending may create a long-lasting cycle (i.e., determining the duration of an expansion cycle followed by a recession). Following the contributions of [Burns and Mitchell \[1946\]](#), cycle analysis shifted towards disentangling time series into cyclical versus trend components. This change in cycle analysis also resulted in some theoretical interpretations of the cycle through the establishment of short-run and long-run macroeconomics (fluctuations vs. growth). These materialize through new theories of economic growth (e.g., [Harrod \[1948\]](#), [Solow \[1956\]](#)) and theories of fluctuations (e.g., [Keynes \[1937\]](#), [Samuelson \[1948\]](#)). In this lecture, we will focus solely on the business cycle theory which emerged in the 1970s and 1980s.

²For instance, in the European Union, there has been ongoing debate since the Maastricht Treaty about fiscal rules and public deficits. In order to maintain a balanced government budget, the EU may penalize any member state exhibiting excessive structural deficits. The critical aspect lies in the definition of the 'structural' nature of the deficit, which typically relies on business cycle measures. A change in the computation of the cyclical budget deficit may alter the stance of the EU Commission on national fiscal rules.

1.2 Measuring business cycles

However, the frontier separating short run and long run analysis is however not as clear as what one would expect. The emergence of econometrics in the second half of the 20th century had also been accompanied by new statistical methods that aim at separating low and high components in economic time series. Any macroeconomist working in a time series face the following problem:

$$y_t = \tau_t + \psi_t, \quad (1)$$

where τ_t is the secular and ψ_t the cyclical component of time series y_t . How can we compute τ_t and ψ_t ? Unfortunately, there is no unique method to disentangle τ_t and ψ_t . Therefore, the measure of business cycles is still an open question. In this section, we will discuss three different methods to extract the business cycle component from the data.

A preliminary step before working on real data is to apply logs. Why do we use logs? Because GDP data exhibit an exponential shape. To illustrate this aspect, let us assume for now that economic growth is following an exponential growth process (a very common assumption). Formally, let Y_t denote the GDP that follows the following data-generating process:

$$Y_t = Y_0 (1 + \gamma)^t \exp(\varepsilon_t). \quad (2)$$

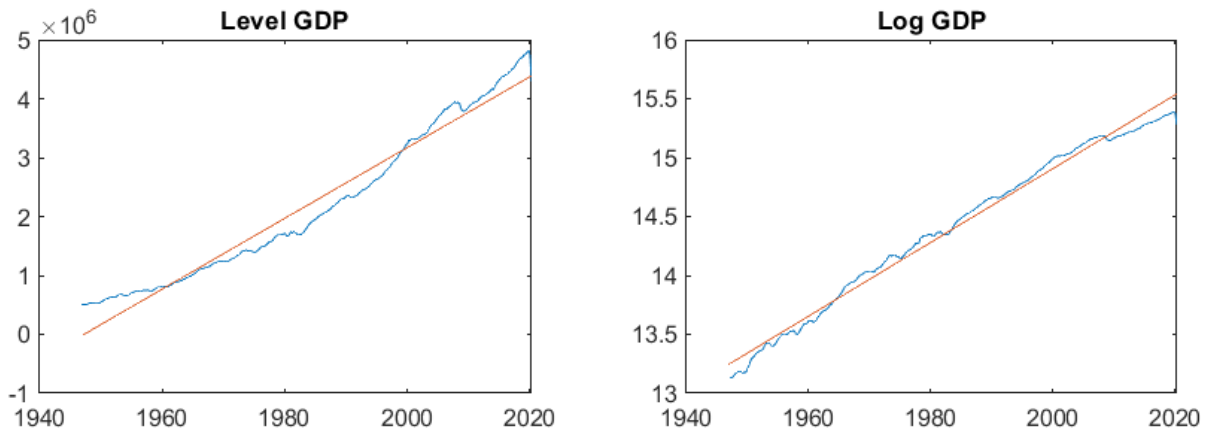
In this expression, $(1 + \gamma)^t$ is the (exponential) trend, ε_t is the fluctuation term and Y_0 is the initial observation of output. In this expression, the interpretation of γ is simply the quarterly percentage growth rate. Applying logs to this expression, we 'linearize' the expression (2) by flattening the exponential shape of output:

$$\log(Y_t) = \log(Y_0) + \log(1 + \gamma)t + \varepsilon_t \quad (3)$$

$$\simeq \underbrace{y_0 + \gamma \cdot t}_{\tau_t} + \underbrace{\varepsilon_t}_{\psi_t}. \quad (4)$$

As the final expression is linear, it is possible to estimate intercept y_0 and slope γ through OLS. From these estimated parameters, the trend can be computed and removed from level output to obtain what we usually interpret as the fluctuation term ψ_t in the GDP time series. See the first MATLAB box to see alternative filtering methods to extract ψ_t from Y_t .

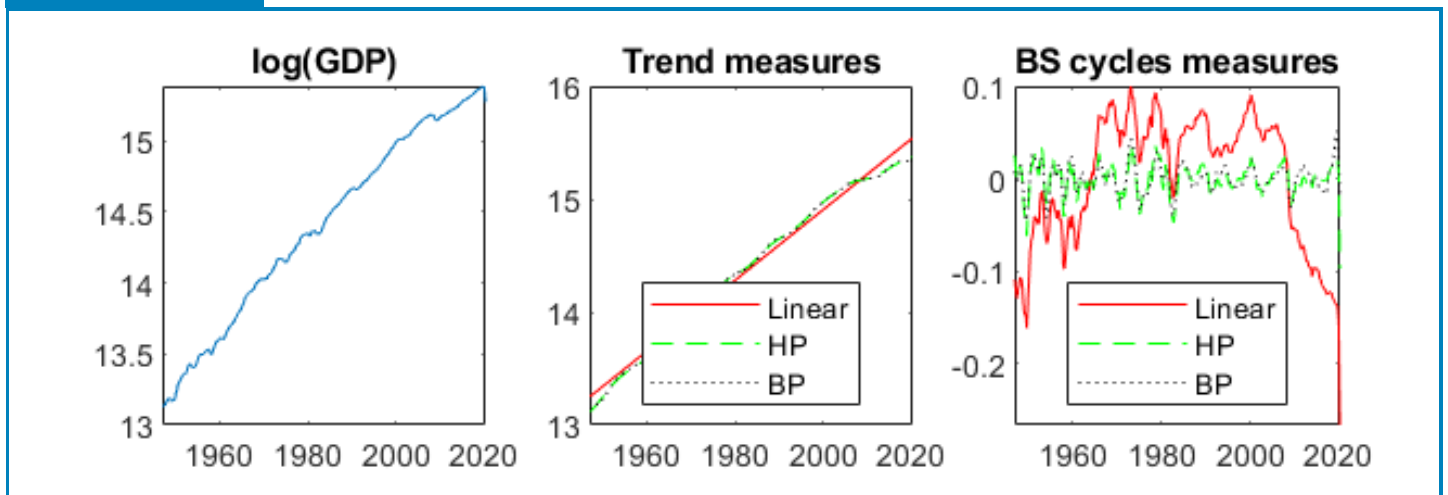
MATLAB box 1 [logGDP.m](#)



We apply three filtering methods on the log of GDP to illustrate some alternative measures of business cycles:

1. **A linear trend:** In a nutshell, we use OLS regression to estimate the slope of growth. This method assumes that the data-generating process is the same as in [Equation 2](#). It allows for the quick extraction of the business cycle component without losing important information in the data. However, if output permanently shifts following a structural break, then the measure of the business cycle may be biased. For example, following the Great Recession in 2007, the US never returned to the pre-crisis trend, resulting in a permanently negative business cycle component after 2007. Considering the potential for structural changes in economic growth, it may be useful to consider a time-varying trend.
2. **The HP filter:** The statistical method of [Hodrick and Prescott \[1997\]](#) extracts the cyclical component of a time series by separating long-term and short-term fluctuations. This method is two-sided, meaning that for computation in period t , it requires both past and future values. Thus, end-of-sample estimations are typically biased. In addition, the filter may omit information in the data that could be relevant for analysis. This method has faced many critiques, summarized by [Hamilton \[2018\]](#).
3. **The band-pass filter:** A band-pass filter is a device that passes frequencies within a certain range and attenuates frequencies outside that range. In economics, the band-pass filter can be used to isolate the component of a time series within a particular frequency band to separate the business cycle, the medium-term cycle, and the long-term cycle (i.e., the trend). This is useful, for instance, to examine if an economic relationship (e.g., money-inflation) remains significant across different frequencies of the cycle. [Baxter and King \[1999\]](#) and [Christiano and Fitzgerald \[2003\]](#) provide the two most renowned applications of the band-pass filter in business cycle analysis.

MATLAB box 2 BSmeasures.m



1.3 The analysis of business cycles

[Blanchard and Watson \[1986\]](#) provides a nice description of modern business cycle analysis: “*Fluctuations in economic activity are seen as the result of small, white noise shocks that affect the economy through a complex dynamic propagation system.*” Pioneered by [Frisch \[1933\]](#), theoretical interpretations of economic fluctuations (i.e. the term ε_t in [Equation 2](#)) rely on system response analysis following the realization of

some shocks. The dominant view is that even large fluctuations result from small shocks.³ Particular economic fluctuations can be ascribed to particularly large shocks followed by periods during which the economy returns to equilibrium. This interpretation contrasts with the “old” Keynesian approach of economic fluctuations: the latter interpret fluctuations as imbalances in money or labor markets.

According to [Lucas Jr \[1977\]](#), the business cycle exhibits some regularities stable over time:

1. The main regularity is at a sectoral level: output movement across sector are strongly serially correlated. So if one sector experiences one boom, it's very likely that all other sector experience also experience a boom too.
2. Production of durable goods is more volatile than non-durable goods.
3. Price are procyclical.
4. Short term interest rates are procyclical.
5. Monetary aggregates and velocity measures are procyclical.

Given these observations, [Lucas Jr \[1977\]](#) argued that “*all business cycle are alike*”. This assertion has important implications for theorists, as it implies that these business cycle regularities can be all interpreted from a single general theory of fluctuations, the real business cycle theory. In the following years, Lucas has provided several critical contributions to the foundation of a theory of business cycles. The main ingredients are a general equilibrium with infinitely-living and representative agents that form their expectations rationally. This handout provides the basis of business cycle theory pioneered by Bob Lucas.

³Some more recent applications to financial turmoils, such as [Kiyotaki and Moore \[1997\]](#) or [Boissay et al. \[2016\]](#), show that small shock can trigger large depressions.

2. An illustration of a business cycle model through the consumption-saving problem

According to [Lucas Jr \[1977\]](#), a theory of real business cycle aims at providing any point in time de value of output. In particular, it provides some theoretical interpretations of the empirical relations across macroeconomic time series, based on agent-level decisions (usually referred to as *microfoundations*). In this section, we present the consumption-saving problem to examine how changes in (expected) consumption affect the equilibrium real rate.

2.1 A toy model of consumption-saving

Consider an simplified version of the [Lucas Jr \[1978\]](#)'s asset pricing model. This model is characterized by discrete-time and an infinite horizon economy populated by *households*, which are infinitely lived and of measure one. This is an endowment economy, so the production is assumed to be fully exogenous and driven by an exogenous shock:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

The production in this economy stems from the presence of infinitively-lived trees, output is the fruit that falls from trees. Therefore, we interpret ε_t as the weather that disrupt the quantity of fruits that the tree can provide (i.e., for a low realization of ε_t , the tree provides a lower quantity of fruits). The weather exhibits some persistence, so a low realization of the weather today is likely to affect the quantity of fruits in the next period by a degree ρ . To illustrate this persistence, a storm may damage a branch, the latter requires few periods to heal and produce fruit as usual. In this model, the only source of uncertainty is about current and future values of ε_t , as the household ignore how much of fruits she will have. This uncertainty can be addressed by the purchased of some financial claims that we discuss hereafter.

Regarding the preferences, each household maximizes her lifetime utility, measured as the current and future discounted streams of utility:

$$\max_{\{c_t, b_t\}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s u(c_{t+s}), \quad (6)$$

where β is the discount factor.⁴ We impose a logarithm for the utility function $u(c_t) = \log(c_t)$. Expression $\mathbb{E}_t$ denotes the expectation operator. In a rational expectation equilibrium, agents can form expectations about the future states of the economy at no cost. However in a stochastic environment, the expectation of $\mathbb{E}_t \{X_{t+1}\}$ is assessed with all information available in t to agents, meaning that any shock that will occurs between t and $t + 1$ will not be anticipated: a gap between the expectation of $\mathbb{E}_t \{X_{t+1}\}$ and its realization X_{t+1} is observed, thus creating a source of fluctuations.

Regarding the financial aspect of the model, we depart from [Lucas Jr \[1978\]](#) to introduce a (risk free) financial claim in the model. Given the uncertainty stemming from the weather, the household may be interested in purchasing a financial claim to hedge against low realizations of the weather. Put differently, the household may choose to consume all the fruits today, or postpone consumption to the next period by depositing the fruits to a bank. The bank is perfectly competitive, and provide (fruit) deposit services and provide an interest rate r_t for each unit of fruit deposited between t and $t + 1$. The fruit is the numeraire, so the interest rate r_t is directly interpreted in units of fruits.

⁴For $\beta = 1$, the household is perfectly patient and does not discount the future, *i.e.* consuming now or later provides exactly the same utility ($\beta^0 u(c) = \beta^1 u(c) = \beta^2 u(c)$). For any value of $\beta < 1$, the household is impatient, the lower β is, the more impatient the household is ($\beta^0 u(c) > \beta^1 u(c) > \beta^2 u(c)$). Microeconomic evidence suggests that households are impatient, so the discount factor should lie between 0 and 1 (usually close to one).

So each period, the households has to possibility to consume c_t units of fruits or deposit b_t to the bank. The budget constraint faced by the household at any point in time is given by:

$$c_t + b_t \leq b_{t-1}r_{t-1} + y_t, \quad (7)$$

where the left-hand side of the equation gathers the spending of the household, while the right-hand side gather the income from deposits and the production from trees. Note that when the household takes it deposit decisions, she knows the interest rate, r_{t-1} , so there is no uncertainty on the return the household may get from depositing fruits in $t + 1$.

The Lagrangian problem solved by the household is given by:

$$\mathcal{L}_t = \sum_{s=0}^{\infty} \beta^s \left[\log(c_{t+s}) - \lambda_{t+s} (c_{t+s} + b_{t+s} - b_{t-1+s}r_{t-1+s} - y_{t+s}) \right], \quad (8)$$

where λ_t denotes the Lagrangian multiplier over the budget constraint. Two important assumptions are implicitly set on this Lagrangian problem: (i) the budget constraint is also discounted, so the constraint is eternally binding over time (i.e., it binds today and in the future); (ii) the budget constraint binds to equality, so the income is fully spent in [Equation 7](#).⁵

Maximizing [Equation 8](#) in c_t and b_t yield the following first order conditions:

$$(c_t) : \lambda_t = \frac{1}{c_t} \quad (9)$$

$$(b_t) : \lambda_t = \beta E_t \{ \lambda_{t+1} \} r_t \quad (10)$$

Note that if the household consumes and saves by adopting these two equations, she will get the highest feasible level of lifetime utility over the business cycles. We provide an interpretation of these two equations in the following subsection.

2.2 A discussion of the first order conditions

The two first order conditions ([9-10](#)) are one of the main ingredient of real business cycle theory: the [Euler equation](#).

o Marginal utility of consumption (MUC).

In equilibrium, the Lagrangian multiplier over the budget constraint is equal to the marginal utility of consumption: $\lambda_t = u'(c_t)$. The functional form of the utility function $u(c_t)$ critically matters to explain the saving-consumption nexus. So this section provides some insights about the interpretation of the utility function.

Consider an exogenous consumption cycle as defined in [Equation 5](#), what are the implications of this cycle in terms of utility function $u(c_t)$ and marginal utility of consumption λ_t for the household? [Figure 1](#) reports some illustrative simulations of consumption, and their corresponding streams of utility and marginal utility.

When consumption is high in period 3, the individual utility also increases with the level of consumption, so there is a perfect co-movement between utility and consumption. In contrast, the marginal utility exhibits a different pattern of fluctuations with respect to consumption. When consumption is high, the

⁵Note that we can check whether the constraint always binds to equality by tracking the non-negativity of the Lagrangian multiplier.

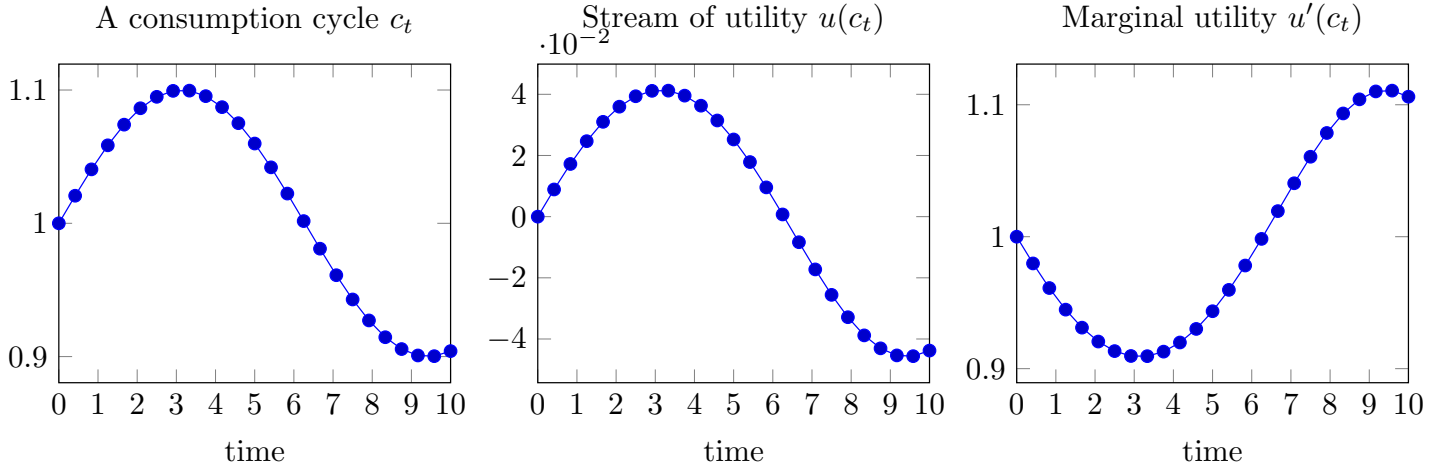


Figure 1: Consumption, utility and marginal utility cycles under logarithmic utility

willingness to consume is low. Why? Because of **diminishing marginal utility**, the consumption of each additional unit of a good gives less extra utility than the consumption of the previous unit. Think about a cyclist running the Tour de France who is thirsty following some physical effort to run up a hill. At first, he's really willing to drink water (λ_t is high), so the first glass of water provides a lot of utility. For the second glass, he becomes less thirsty so the consumption of the second glass does provide utility but not as much as the previous one. This feature of the marginal utility explains the negative relation between consumption and the willingness to consume. When consumption is low, the utility from consuming an additional unit of goods is higher than when consumption is high.

o Intertemporal Marginal Rate of Substitution (IMRS).

The second core key ingredient of modern business cycle theory is the Euler equation. Consider optimal condition on bonds:

$$u'(c_t) = \beta E_t \{u'(c_{t+1})\} r_t. \quad (11)$$

The left hand side of this equation provides the utility loss of saving today (priced in terms of present marginal utility), and the right hand side is the utility gain at the next period. As long as this equation holds to equality, there is a no-arbitrage condition that makes the household equally well off between consuming today or tomorrow. Note that the equilibrium interest rate plays here a key role as it directly determines how households are weighting consumption today versus tomorrow: if the equilibrium rate rises, households are more likely to postpone consumption and to consume more at the next period. Therefore, the key ingredient here is the pricing scheme that affects the trade-off between consuming now and tomorrow. The intertemporal marginal rate of substitution of future for present consumption is given by $u'(c_t)/E_t \{u'(c_{t+1})\}$.

The intertemporal analysis of consumption is actually exactly the same as in the standard consumer problem describing the maximization of the utility by determining the optimal quantity of two types of goods under a budget constraint. The difference here is that instead of having two different goods, the trade-off now applies between consuming today and tomorrow. The budget constraint becomes intertemporal and shapes the dynamics of the interest rate.

Figure 2 reports the joint-determination of the optimal level of financial assets and consumption. In the left panel, we show the consumption possibilities (c_t, c_{t+1}) which are offered by financial markets in

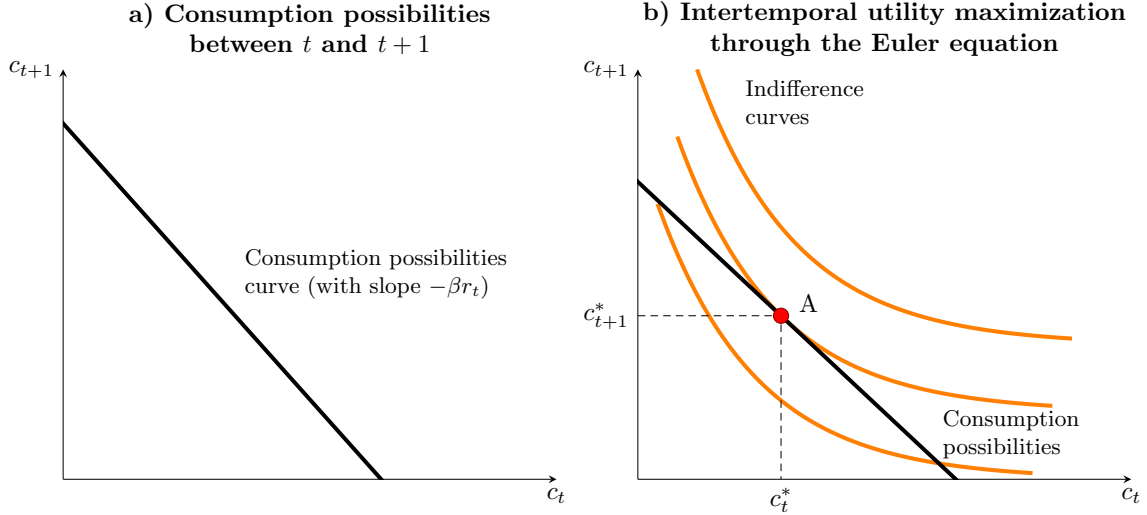


Figure 2: Optimization scheme for the representative household

the economy (and implicitly here by the production capacities of the economy which are exogenous). The slope of this curve, $-\beta \mathbb{E}_t \{r_{t+1}\}$, is decreasing because current and future consumption are substitutes.

In the right panel of Figure 2, we introduce the household's preference with respect to consumption. These preferences materialize in a graphical representation as indifference curves. These curves are different pairs (c_t, c_{t+1}) that provides exactly the same utility so that an individual is equally well off in any point of the indifference curve. Given the real rate on deposits, the household determines the levels of present and future consumption (c_t^*, c_{t+1}^*) maximizing her utility given the consumption possibilities offered by financial markets. In equilibrium A, the household find the unique possible path of consumption maximizing her utility intertemporally and is called *the consumption point*. Equation 11 is actually the (unique) consumption point that provides the highest level of lifetime utility for a given budget constraint.

2.3 General Equilibrium

Finally, we need to set one condition over deposits to close the model. In financial autarky (*i.e.*, with no trading partners), the general equilibrium is jointly determined by the equilibrium on financial and goods markets, as the optimal condition of saving determine the amount of consumption. So clearing one market also clears the others.

In absence of external borrowing/lending, at a macroeconomic level, the following condition on domestic saving emerges:

$$b_t = 0. \quad (12)$$

Note that since households populating our economy are all symmetric, the agent specific saving is also zero in equilibrium. Given this relation, the budget constraint in Equation 7 becomes:

$$c_t = y_t. \quad (13)$$

Therefore, the model can be cast as the following set of two equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \beta^{-1} \mathbb{E}_t \{y_{t+1}\} / y_t \end{cases}. \quad (14)$$

The first equation is the quantity of fruit falling from trees, and the second equation is the equilibrium condition on deposits. In absence of endogenous production, the number of fruit over time determines the equilibrium interest rate that ensure that Equation 12 holds to equality.

Note that in this simple setup, the household does not optimally choose consumption, because it is exogenous (falling from trees). As a result, the consumption-saving is not really a saving problem because saving is zero in equilibrium. So the interpretation of the model is the following: the agent in t contemplates how much fruits y_t she has by observing ε_t (i.e. the weather). Given this new information, the agent plans how much fruits y_{t+1} he may expect at the next period. Given the current and expected levels of consumption, the equilibrium rate adjusts in order to clear financial markets, i.e. the real rate ensures that saving remains zero in equilibrium.

2.4 Expectations

How are we supposed to cope with the term $E_t \{y_{t+1}\}$ in the Euler equation from system (14)? This question is still left open in macroeconomics, as there is no perfect specification for the expectation scheme that portrays sufficiently well the formation process of expectations of agents over time.

Back in the 60s, macroeconomic analysis was relying on large-scale models based on old Keynesian economics. These models were 'backward looking', in a sense that they did not feature any type of expectation in their modeling structure. Some purely forward looking dynamics were simply omitted, which questioned their empirical relevance for policy analysis as argued by Lucas et al. [1976] in his famous critique.

Which forward looking dynamics are we talking about? Let us think about some simple examples in macroeconomics:

1. A first example is about **price setting**. If the central bank is accommodative today, a worker may expect that inflation will rise in the future, so the worker asks today for an increase of her wage to compensate future inflation (which in turn self-fulfills the increase in inflation now). These considerations led to a new interpretation of the Keynesian Phillips curve.
2. Another example concerns **portfolio allocation in open economies**. Suppose that a central bank in the domestic country raises its interest rate. Investing in the domestic country becomes more profitable, so capital is expected to flow into the country and the domestic currency is expected to appreciate in the future. As a result, the domestic currency appreciates immediately in response to the increase in the domestic real interest rate, before the capital inflow. This narrative describes the uncovered interest rate parity condition. These two examples are simply not incorporated in purely backward models.

Back to our problem of having an expectation scheme, Rational Expectations (RE) hypothesis has been the most popular tool to solve current macroeconomic model. RE provides a quick, simple and agnostic way of modelling expectations in macroeconomic models. In addition, RE are typically 'easy' to compute, as you even get closed-form expressions. The concept of rational expectations is pioneered by Muth [1961], an agent having rational expectation **use all relevant information** in forming expectations of economic variables. In our toy model, what is the relevant information for $E_t \{y_{t+1}\}$? Another implicit facet of RE lies in assumption that agents also have perfect knowledge of the model's equation. So if our agent knows the endowment in Equation 5, by iterating forward this equation we get:

$$\begin{aligned} E_t \{y_{t+1}\} &= E_t \{1 - \rho + \rho y_t + \varepsilon_{t+1}\} \\ &= E_t \{1 - \rho + \rho y_t\} \\ &= 1 - \rho + \rho y_t, \end{aligned}$$

So as you can see, we are able to transform $E_t \{y_{t+1}\}$ into a system of current and backward variables. Implicitly, you may have noticed that we imposed $E_t \{\varepsilon_{t+1}\} = 0$. This is because the agent ignores the future realizations of the shock, but know its distribution. Since on average, $E(\varepsilon_t) = 0$, then it is rational to expect that – on average – the shocks are 0. So agent's expectations may be wrong, but **are correct on average** over time.

So, forecast errors are non-zero:

$$E_t \{y_{t+1}\} - y_{t+1} \neq 0 \text{ for } t > 0$$

But converge to zero asymptotically:

$$E \left(E_t \{y_{t+1}\} - y_{t+1} \right)_{t=0}^{+\infty} = 0$$

Back to our two-equation system in [Equation 14](#), under rational expectations, the model becomes:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases} . \quad (15)$$

Given this system, we are able to simulate our model because as long as we know the value of output, we can obtain the value of the real interest rate.

Exercise 1: Simulating the non-linear model

Consider the system in [Equation 15](#).

1. Write a MATLAB file that simulates this system over 400 periods. Use the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
 - (a) Set the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
 - (b) Compute the steady state of the model.
 - (c) Draw 400 exogenous realizations of ε_t from a Gaussian distribution using function `randn()`.
 - (d) Declare an empty vector using function `nan()` for `y` and `r` that we will fill next.
 - (e) Run a loop that fills recursively the vector for `y` and `r`.
2. Plot the simulation.
3. Compute the standard deviation of output and its correlation. Compare with the empirical data of real output that we have computed using a linear trend.
4. If you find a gap, adjust the calibration of σ and ρ to match US data.
5. Simulate an Impulse Response Function (IRF) of the model.
6. Does a large output expansion has the same effect on the real rate than a large depression? Compare the absolute values of the IRFs.
7. Plot the histograms of your simulated data and explain why the asymmetry is different for the distribution of output and the real rate.

2.5 A discussion on stochastic uncertainty

Note that in [Equation 15](#), our solution is fully consistent with rational expectations, as we analyze a case in which risk aversion is implicitly one. Consider a more general CRRA specification, $u(c) = 1/(1-\sigma)(c^{1-\sigma} - 1)$, with $u'(c) = c^{-\sigma}$

$$\begin{aligned} r_t &= \beta^{-1} E_t \{ y_{t+1}^\sigma \} / y_t^\sigma \\ &= \beta^{-1} E_t \{ (1 - \rho + \rho y_t + \varepsilon_{t+1})^\sigma \} / y_t^\sigma \end{aligned}$$

In this expression, $y_t \sim N(E(y_t), V(y_t))$,

3. Approximation methods

In the two-dimensional model considered, it is possible to simulate the (fully) non-linear model without approximating the equations. However, if the model gets more complicated (in particular with state variables), it's not possible to get a closed-form expression of the non-linear model. Therefore, approximation methods are applied to the model's equation to obtain a closed-form expression of a linear transition equation.

3.1 The problem

Our 2 equation model is a non-linear system with rational expectations. In compact notation, the system can be represented as follows:

$$\mathbb{E}_t [f_\theta (x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0 \quad (16)$$

where:

- x is the vector collecting n endogenous variables;
- ϵ is the vector collecting the p exogenous stochastic shocks;
- θ is the vector collecting the deep parameters of the model;
- $f_\theta(\cdot)$ is the equations of the model;

The solution to a model like this is normally a set of functions relating controls to states such that our 2 equations are satisfied at all times (i.e. in $t+1$, t and $t-1$). The solution of the system x_t takes the form a decision rule:

$$x_t = g(x_{t-1}, \epsilon_t), \quad (17)$$

where function $g(\cdot)$ is unknown, but theoretically can be computed from [Equation 16](#). Let us first rewrite the system in [Equation 16](#) by injecting the model's solution in [Equation 17](#). Recall, that under rational expectations, $E_t\{x_{t+1}\} = g(x_t, \epsilon_{t+1})$ where $E(\epsilon_{t+1}) = 0_p$. To get rid off of the expectation term in the previous expression, we obtain:

$$f_\theta(g(x_t, 0_p), x_t, y_{t-1}, \epsilon_t) = 0. \quad (18)$$

Iterating again to express our system as a function of x_{t-1} :

$$f_\theta(g(g(x_{t-1}, \epsilon_t), 0_p), g(x_{t-1}, \epsilon_t), x_{t-1}, \epsilon_t) = 0 \quad (19)$$

Therefore, the true solution of the system [Equation 16](#) is a function $g(\cdot)$ that satisfies [Equation 19](#). The problem is that, in case of larger models, there is an infinite number of values of y_{t-1} and ϵ_t and an infinite number of values of y_t to find. Hence, except for few exceptions, the usual practice is to approximate $g(\cdot)$.

Note that in our two-dimensional model of consumption-saving, we are able to get a closed-form expression $g(\cdot)$ but in more complicated models, it's impossible to obtain the true value of $g(\cdot)$. The next subsection aims at explaining how we can approximate $g(\cdot)$.

The closed-form expression of the policy rule $x_t = g(x_{t-1}, \epsilon_t)$ is given by:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \quad (20)$$

$$r_t = \frac{\rho}{\beta} + \frac{\beta^{-1}(1 - \rho)}{1 - \rho + \rho y_{t-1} + \varepsilon_t}. \quad (21)$$

3.2 Linearization Method

The standard way to bound the problem is to approximate the function $g(\cdot)$ by a log-linear function around the steady state. There are many different solution methods (e.g. Schur decomposition, undetermined coefficients, etc) to reach the linearized version of $g(\cdot)$. Under particular conditions defined by [Blanchard and Kahn \[1980\]](#), our linear rational expectations model has a stable and convergent intertemporal equilibrium.

More precisely, we usually take the following steps to solve a model:

1. Find the steady state of the model;
2. (log-)Linearize around the steady state;
3. Write the Dynare file.

For any function $f(x_t)$ with corresponding steady state $f(\bar{x})$, a Taylor expansion up to first order is given by:

$$f(x_t) \simeq f(\bar{x}) + f'(\bar{x})(x_t - \bar{x}) \quad (22)$$

Expressed in terms of deviation from the steady state, $f(x_t) - f(\bar{x})$, the expansion can be rewritten in the following way:

$$\begin{aligned} f(x_t) - f(\bar{x}) &\simeq f'(\bar{x})(x_t - \bar{x}) \\ &\simeq f'(\bar{x})\bar{x} \frac{(x_t - \bar{x})}{\bar{x}} \\ &\simeq f'(\bar{x})\bar{x} \log\left(\frac{x_t}{\bar{x}}\right) \\ &\simeq f'(\bar{x})\bar{x} \hat{x}_t \end{aligned}$$

where $\hat{x}_t = \log(x_t/\bar{x})$ is the percentage variations of x_t around its steady state.

For a two-variable function, the approximation reads as:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})\bar{x}\hat{x}_t + f'_y(\bar{x}, \bar{y})\bar{y}\hat{y}_t \quad (23)$$

We can then summarize these general approximations with a few rules that we can use to log-linearize quickly a model:

Level variable	Log-linearized variable
x_t	$\bar{x}\hat{x}_t$
$x_t y_t$	$\bar{x}\bar{y}(\hat{x}_t + \hat{y}_t)$
$\frac{x_t}{y_t}$	$\frac{\bar{x}}{\bar{y}}(\hat{x}_t - \hat{y}_t)$
x_t^a	$\bar{x}^a a \hat{x}_t$
$\log(x_t)$	$\hat{x}_t$
$\exp(x_t)$	$\exp(\bar{x})\bar{x}\hat{x}_t$

3.3 Linearization application

Let us now linearize our system of equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases} \quad (24)$$

Steady state. A preliminary step before performing the approximation to our model's equations is to characterize the steady state of the model (i.e. the fixed point of the approximation). Following the linearization, each variable is expressed in percentage deviation from the steady state. To obtain the steady state, one just needs to take off the time subscript on endogenous variables:

$$\begin{cases} \bar{y} = 1 - \rho + \rho\bar{y} \\ \bar{r} = \frac{(1-\rho+\rho\bar{y})}{\beta\bar{y}} \end{cases} \quad (25)$$

Note that the following system in the (deterministic) steady state reads as follows:

$$\begin{cases} \bar{y} = 1 \\ \bar{r} = \frac{1}{\beta} \end{cases} \quad (26)$$

Linearizing the equations. Next, we must linearize the equations of the model. We apply here:

$$\underbrace{y_t}_{\bar{y}\hat{y}_t} = \underbrace{1 - \rho}_0 + \underbrace{\rho y_{t-1}}_{\rho\bar{y}\hat{y}_{t-1}} + \underbrace{\varepsilon_t}_{\varepsilon_t}$$

Note that since $\bar{y} = 1$, then equation boils down to:

$$\hat{y}_t \simeq \rho\hat{y}_{t-1} + \varepsilon_t$$

Some remarks on the linearization. You may struggle to log-linearize, but with learning-by-doing, you will get used to it. After some time of practice, linearizing equations get very quickly. Note also that the stochastic shock is not affected by the linearization process

$$\begin{aligned} r_t &= \frac{(1 - \rho + \rho y_t)}{\beta y_t} \\ r_t &= \frac{(1 - \rho)}{\beta} \frac{1}{y_t} + \frac{\rho}{\beta} \\ \bar{r}\hat{r}_t &\simeq -\frac{1 - \rho}{\beta\bar{y}}\hat{y}_t \\ \hat{r}_t &\simeq -(1 - \rho)\hat{y}_t \end{aligned}$$

So the linerarized system reads as:

$$\begin{aligned} \hat{y}_t &\simeq \rho\hat{y}_{t-1} + \varepsilon_t \\ \hat{r}_t &\simeq -(1 - \rho)\hat{y}_t \end{aligned}$$

And we are looking for a solution of the form for $\hat{x}_t = [\hat{y}_t, \hat{r}_t]$:

$$\hat{x}_t = P\hat{x}_{t-1} + Q\varepsilon_t$$

While equation for $\hat{y}_t$ is recursive, the other in $\hat{r}_t$ is not. We simply inject $\hat{y}_t$ in $\hat{r}_t$ to force recursion: $\hat{r}_t \simeq -(1 - \rho)\hat{y}_t = -(1 - \rho)\rho\hat{y}_{t-1} - (1 - \rho)\varepsilon_t$. The solution of the system reads as:

$$\underbrace{\begin{bmatrix} \hat{y}_t \\ \hat{r}_t \end{bmatrix}}_{\hat{x}_t} = \underbrace{\begin{bmatrix} \rho & 0 \\ -(1 - \rho)\rho & 0 \end{bmatrix}}_P \underbrace{\begin{bmatrix} \hat{y}_{t-1} \\ \hat{r}_{t-1} \end{bmatrix}}_{\hat{x}_{t-1}} + \underbrace{\begin{bmatrix} 1 \\ -(1 - \rho) \end{bmatrix}}_Q \varepsilon_t$$

As long as we know vector ε_t , matrices P and Q , and initial conditions $\hat{y}_0$ and $\hat{r}_0$, we are able to characterize the path of $\{\hat{y}_t, \hat{r}_t\}_{t=1}^\infty$.

Exercise 2: Simulating the linear model

Consider now the linear system. This exercise can be coded in the same script file as Exercise 1.

1. Write a MATLAB file that simulates this system over 400 periods.
2. Plot the simulation.
3. Simulate an Impulse Response Function (IRF) of the model. An IRF is just the same simulation as before, but with one shock at the first period and next no shocks. It isolates the pure propagation mechanism.
4. Does a large output expansion has the same effect on the real rate than a large depression? Compare the absolute values of the IRFs.
5. Plot the histograms of your simulated data and explain why the asymmetry is different for the distribution of output and the real rate.
6. Compute the Errors of approximation on the nominal rate $E[(r_t - \hat{r}_t)^2]$.

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